

Call for Expressions of Interest

<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	Personal Neurotech App for Athletes
<i>Company:</i>	Aelo Neurotech

Contact Details

<i>Lead Contact:</i>	Max L. Wilson
<i>Lead E-mail:</i>	Max.wilson@nottingham.ac.uk
<i>Additional Contact:</i>	Charlotte Keatch
<i>E-mail:</i>	charlotte@aelo.com
<i>Additional Contact:</i>	Additional contact name
<i>E-mail:</i>	Additional contact e-mail
<i>Additional Contact:</i>	Additional contact name
<i>E-mail:</i>	Additional contact e-mail

Additional Info

<i>Students must sign NDA:</i>	☒
<i>Project uses personal data:</i>	☒
<i>Custom hardware required:</i>	☐ If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	Human-Computer Interaction, Brain Data Group

Project Background Information

Aelo Neurotech make a consumer neurotech device for athletes. The aim is to provide athletes with the ability to record and reflect on brain data as part of their training and to reflect on performance while competing. In rowing, for example, there is focus on training days on a rowing machine or on the water. During events, there is mental preparation on the bank before competing, there is often a long delay between getting on the water and being at the start line where people need to remain focused, and then there is the focus people retain during the race. There is also a consideration between the mental activity of an individual or a whole crew working together in synchrony. The company has built various levels of prototypes that will become available during the year. We have alternative hardware as a fallback.

The aim of this project is to design and build the app, focusing on when it should be used and how it should visualise the relationships between the data points to provide value to an athlete.

The device is built into caps and other sports-headwear, and is designed for continuous measurement across a whole day. Neurotech typically works best when it can get moments of stillness, but can also be worn during movement if the movement is predictable. Rowing is a possible case study, but students

¹ An overview of different research areas in the school of computer science can be found [here](#).

can propose a viable alternative sport. Collecting data about people at work and studying is a fallback.

The aim is to generate ideas for the final product app, and inspire new thought for the company. The aim of the work is more focused on conveying meaningful insight from their cognitive metrics to athletes, rather than on "being pretty".

Requirements

The group must provide:

- 1) An analysis of the different use cases in a selected sport
- 2) Multiple prototypes of the app, to compare how they can be used
- 3) A working final prototype app, likely built for android

Optionally, groups can make advances on processing the brain data (fNIRS), but the focus is on the design and build for a given use case.

Skills

Highly desirable:

- *Experience in a relevant sport*
- *Mobile development*
- *Design and prototyping*

Desirable:

- Prior UX experience
- Javascript full stack
- AI/ML experience with physiological sensor data

Submission / Award information

<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.